College Physics 2

Fluid Mechanics

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College Physics 2

Density and Pressure

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College Physics 2

Archimedes' Principle

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Density and Pressure

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OBJECTIVES

- Define and apply the concepts of density and fluid pressure to solve physical problems.
- Define and apply concepts of absolute, gauge, and atmospheric pressures.

DEFINITION OF TERMS

FLUID MECHANICS is the branch of science which deals with the study of the behavior of fluids (gases or liquids)

FLUID is any substance that can flow either a liquid or a gas

FLUID STATICS is the study of fluids at rest or equilibrium situations

FLUID DYNAMICS is the study of fluids in motion.

DEFINITION OF TERMS

DENSITY is defined as mass per unit volume.

A homogenous material such as iron or ice has some density ρ throughout.

$$\boldsymbol{\rho = \frac{m}{V}}$$

Where:

m = mass (kg)

V = volume (m^3)

Figure 1. Two objects with different masses and different volumes but the same density. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp373) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

DENSITIES OF SOME COMMON SUBSTANCES

Material	Density (kg/m^3)*	Material	Density (kg/m^3)*
Air (1 atm, 20°C)	1.20	Iron, steel	7.8×10^3
Ethanol	0.81×10^3	Brass	8.6×10^3
Benzene	0.90×10^3	Copper	8.9×10^3
Ice	0.92×10^3	Silver	10.5×10^3
Water	1.00×10^3	Lead	11.3×10^3
Seawater	1.03×10^3	Mercury	13.6×10^3
Blood	1.06×10^3	Gold	19.3×10^3
Glycerine	1.26×10^3	Platinum	21.4×10^3
Concrete	2×10^3	White dwarf star	10^{10}
Aluminum	2.7×10^3	Neutron star	10^{18}

*To obtain the densities in grams per cubic centimeter, simply divide by 10^3.

Table 1. Densities of Some Common Substances. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp374) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

RELATIVE DENSITY

The relative density ρ_r of a material is the ratio of its density to the density of water (1000 kg/m^3).

$$\rho_r = \frac{\rho_x}{1000\ \text{kg/m}^3}$$

Steel (7800 kg/m^3)	$\rho_r = 7.80$
Brass (8700 kg/m^3)	$\rho_r = 8.70$
Wood (500 kg/m^3)	$\rho_r = 0.500$

SAMPLE

1. Find the mass and weight of the air at 200C in a living room with a 4.0 m by 5.0 m and a ceiling at 3.0 m high.

Solution

$$V = (4.0\text{m})(5.0\text{m})(3.0) = 60\ \text{m}^3$$

$$m_{air} = \rho_{air}V = (1.20\ \text{kg/m}^3)(60\ \text{m}^3)$$

$$m_{air} = 72\ \text{kg}$$

$$w_{air} = m_{air}g = (72\text{kg})(9.81\text{m/s}^2) = 706.32\ \text{N}$$

$$w_{air} = 706.32\ \text{N}$$

PRESSURE

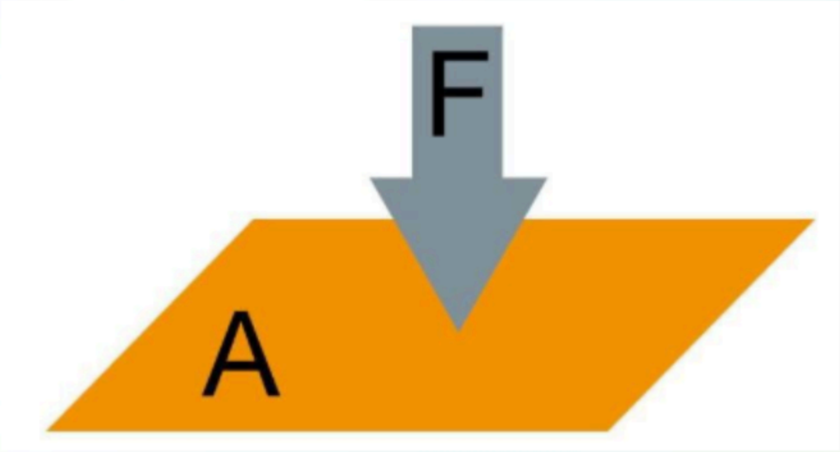
It is the amount of force exerted on a given area. Its SI unit is Pascal (Pa)

$$P = \frac{F}{A}$$

Where:

P = pressure (Pa)
F = force in Newton (N)
A = area (m^2)

Unit: Pascal (Pa)
1Pa = 1 N/m^2



If the applied force is acting on a small area, then the pressure will be large.

Figure 2 Pressure Formula . From. *https://www.testo.com/en-IN/services/knowledgebase-pressure-physical-principles*

Fluid Pressure

Suppose we have a tank filled with fluid with density ρ.
The tank is filled up to a height "h" with a cross sectional area "A"

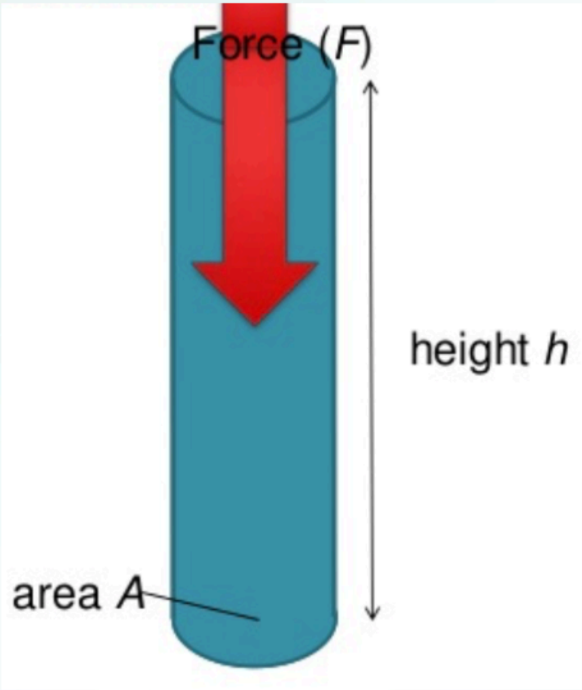


Figure 3. Tank filled with fluid. From. *https://www.slideshare.net/reastment/pressure-in-fluids*

$$P = \rho g h$$

Where

ρ = density (Pa)
g = gravitational acceleration (m/s^2)
h = height (m)

As you go deeper the swimming pool, the pressure gets higher

FLUID PRESSURE

A liquid or gas cannot sustain a shearing stress - it is only restrained by a boundary. Thus, it will exert a force against and perpendicular to that boundary.

The force F exerted by a fluid on the walls of its container always acts perpendicular to the walls.

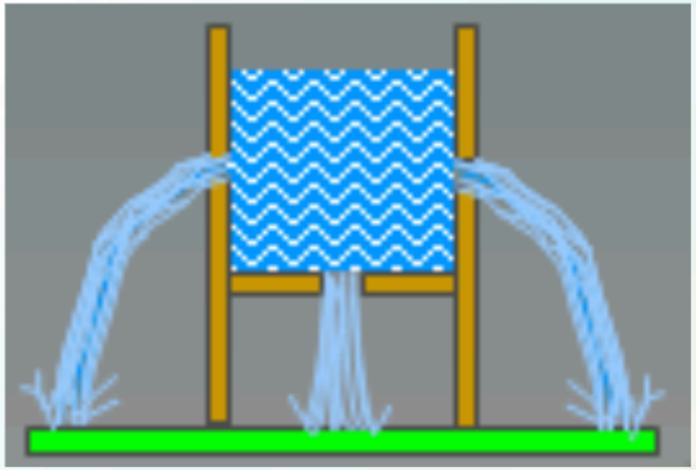


Figure 4. Fluid exerted by a fluid on the walls.. From. *https://www.slideshare.net/nosuhaila/chapter2-24683465*

FLUID PRESSURE

Fluid exerts forces in many directions. Try to submerse a rubber ball in water to see that an upward force acts on the float.

Fluids exert pressure in all directions.

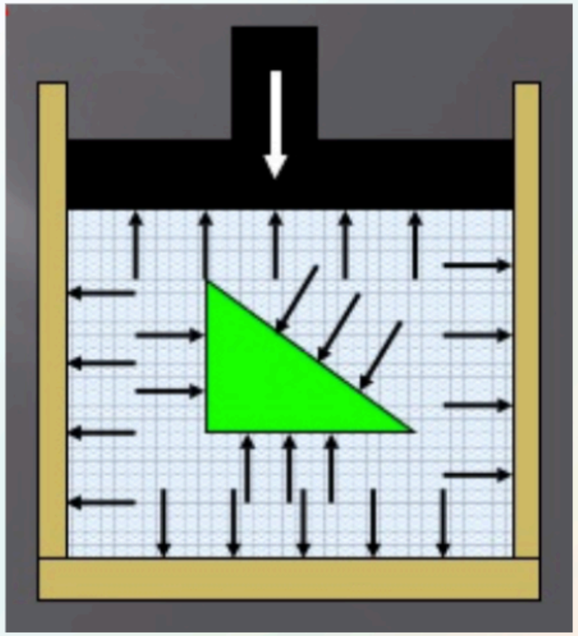


Figure 5. Fluid exerted in all direction. From. *https://www.slideshare.net/nosuhaila/chapter2-24683465*

INDEPENDENCE OF SHAPE AND AREA.

Water seeks its own level, indicating that fluid pressure is independent of area and shape of its container.

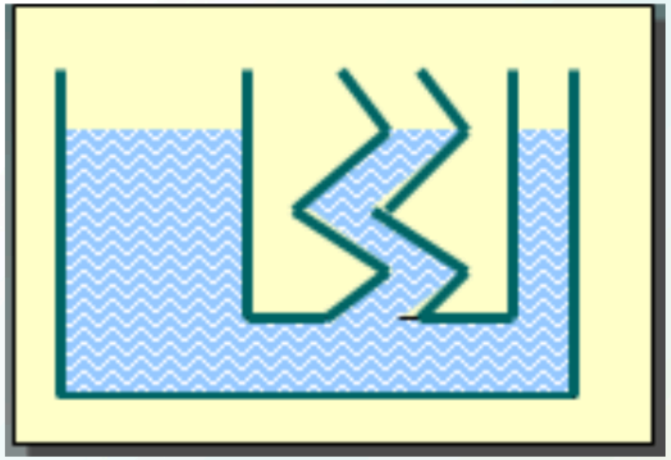


Figure 6. Independent of Shape and Area. From. *https://www.slideshare.net/nosuhaila/chapter2-24683465*

At any depth h below the surface of the water in any column, the pressure P is the same. The shape and area are not factors.

PROPERTIES OF FLUID PRESSURE

- The forces exerted by a fluid on the walls of its container are always perpendicular.
- The fluid pressure is directly proportional to the depth of the fluid and to its density.
- At any particular depth, the fluid pressure is the same in all directions.
- Fluid pressure is independent of the shape or area of its container.

SAMPLE

1. A diver is located 20 m below the surface of a lake (r = 1000 kg/m3). What is the pressure due to the water?

The difference in pressure from the top of the lake to the diver is:

$\Delta h = 20$ m; $g = 9.8$ m/s2

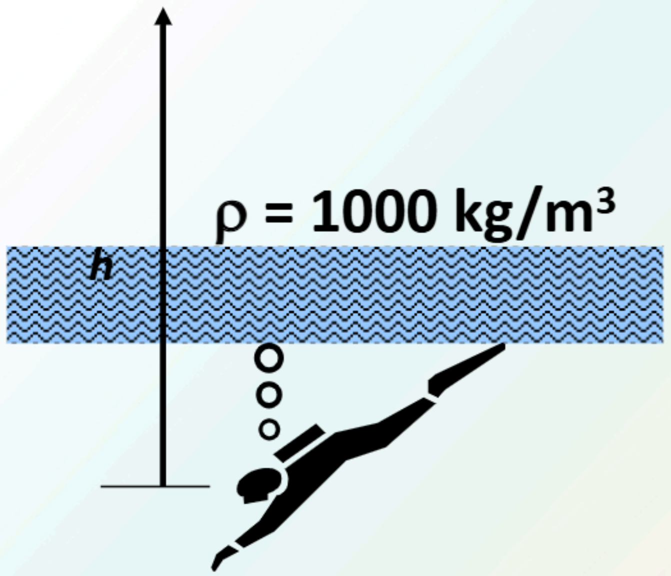


Figure 7. Diver . From. *https://www.slideshare.net/nosuhaila/chapter2-24683465*

$$\Delta P = \rho g \Delta h = (1000\ kg/m^3)(9.8\ m/s^2)(20m)$$
$$= 196{,}000\ Pa$$

ATMOSPHERIC PRESSURE

For an open tank, we should include the pressure outside the tank pressing on the top of the surface of the fluid, which is the atmospheric pressure $\mathbf{P_{atm.}}$

$$P_{\mathbf{atm}} = 101.3 \text{ kPa} = 760 \text{ torr} = 760 \text{ mmHg}$$

$$P_{\mathbf{total}} = P_{\mathbf{atm}} + \boldsymbol{\rho g h}$$

SAMPLE

2. Compare the total pressure at the bottom of a swimming pool of depth 3.00m if it's filled with fresh water and seawater.

SOLUTION

For freshwater

$P_{total} = P_{atm} + \rho gh$

$= 101.3 \times 10^3 \text{ Pa} + (1 \times 10^3 \text{ kg/m}^3)(9.81)(3.0\text{m})$

$= 130{,}730 \text{ Pa}$

For saltwater

$P_{total} = P_{atm} + \rho gh$

$= 101.3 \times 10^3 \text{ Pa} + (1.03 \times 10^3 \text{ kg/m}^3)(9.81)(3.0\text{m})$

$= 131{,}612.9 \text{ Pa}$

CONVERSION OF PRESSURE UNITS

Name of unit	Unit	Conversion
Pascal	Pa	1 Pa = 1 N/m^2
Bar	bar	1 bar = 0.1 MPa
Water column metre	mH_2O	1 mH_2O = 9 806.65 Pa
Atmospheric pressure	atm	1 atm = 101 325 Pa
Mercury column metre	mHg	1 mHg = 1/0.76 atm
Torr	torr	1 torr = 1 mm Hg

HYDRAULIC PRESSURE

Consider a hydraulic jack filled with fluid everywhere.

Pascal's principle states that "an external pressure applied to an enclosed fluid is transmitted uniformly throughout the volume of the liquid."

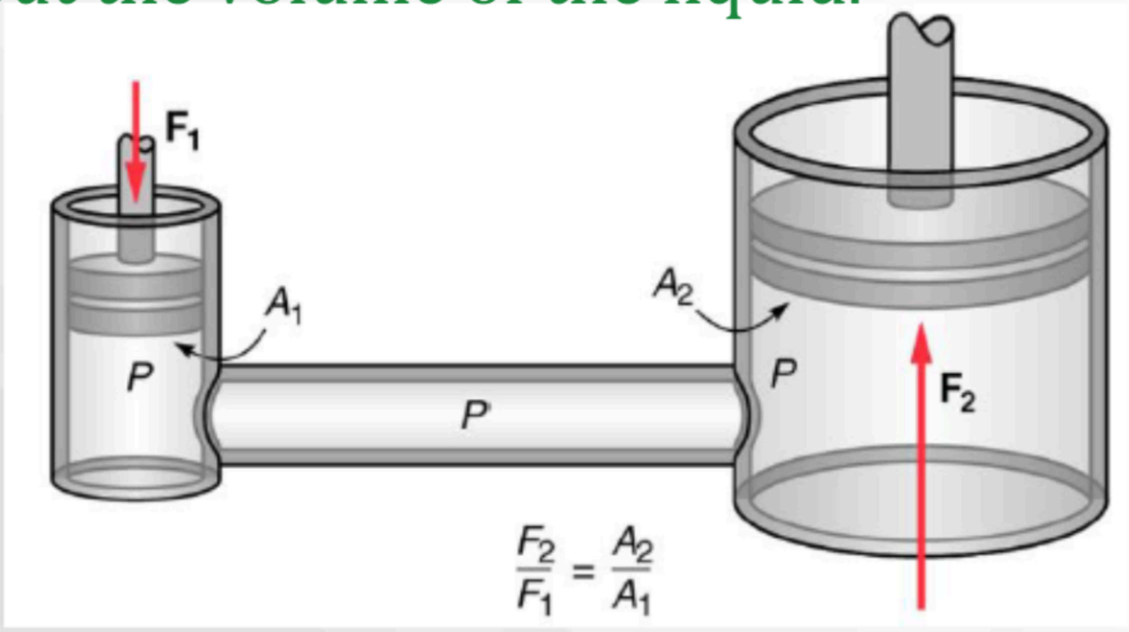


Figure 8. Hydraulic Jack Filled with Fluid. From. *https://www.chegg.com/homework-help/questions-and-answers/pascal-s-law-principle-transmission-fluid-pressure-states-pressure-exerted-anywhere-confin-q3033100*

Pressure in = Pressure out

HYDRAULIC PRESSURE

The volume also are equal. Supposed pushing it with a distance d_1, on the left , then the amount of fluid displaced is the volume that that fluid that ends up at the other side d_2.

In other words, the same amount of fluid leaves on the left and enters on the right side.

$$\mathbf{A_1 d_1 = A_2 d_2}$$

SAMPLE

3. A car that weighs 10.0 kN is placed on the 1.00 m radius piston of a hydraulic press. How much force should be exerted in 5.0 cm radius piston to lift the car?

Solution:

$$\frac{F_{in}}{A_{in}} = \frac{F_{out}}{A_{out}}$$

$$F_{in} = \frac{A_{in}F_{out}}{A_{out}}$$

$$F_{in} = \frac{\pi(0.05m)^2(10000N)}{\pi(1m)^2} = 25N$$

Answer: 25 N

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Physics book by Gary B. Conrado and Maria A. Boncodin.

Serway, R. A., & Vuille, C. (2015). College Physics 10th Edition. Stamford, Connecticut: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

REFERENCES

https://www.slideshare.net/nosuhaila/chapter2-24683465

https://www.slideshare.net/reastment/pressure-in-fluids

https://www.testo.com/en-IN/services/knowledgebase-pressure-physical-principles

https://www.chegg.com/homework-help/questions-and-answers/pascal-s-law-principle-transmission-fluid-pressure-states-pressure-exerted-anywhere-confin-q3033100

College Physics 2

Archimedes' Principle

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OBJECTIVES

- Define Buoyant Force
- State Archimedes' Principle
- Solve physical problems involving Archimedes' Principle

Buoyant Force

Have you ever tried to push a ball down under water?

It is extremely difficult to do because of large upward force exerted by the water on the ball.

This upward force is what we call Buoyant force.

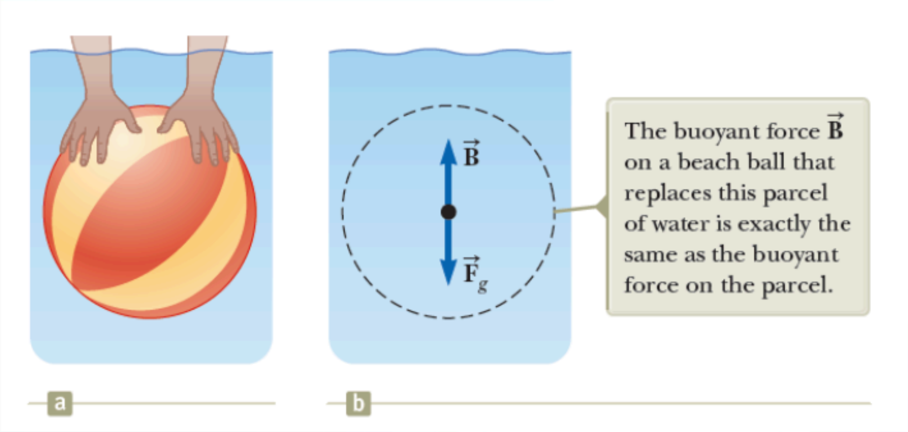


Figure 1. a) A swimmer pushes a beach ball under water. b) The forces on a beach ball–sized parcel of water. From. *Physics for Scientists and Engineers with Modern Physics*(pp 424) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

Archimedes' Principle

"The magnitude of the buoyant force on an object always equals the weight of the fluid displaced by the object."

Figure 2. Archimedes From. *Physics for Scientists and Engineers with Modern Physics*(pp 424) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

Archimedes
Greek Mathematician, Physicist, and Engineer (c. 287–212 BC)

Archimedes' Principle

An object that is completely or partially submerged in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced.

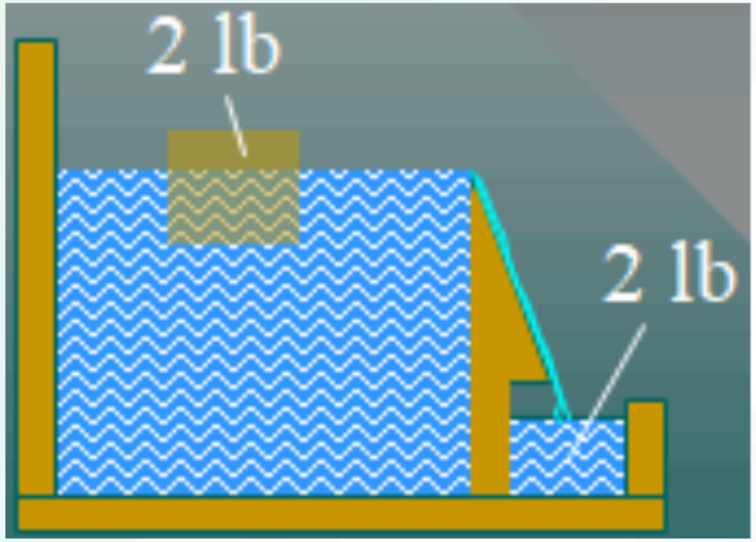


Figure 3. Buoyant Force. From. *https://www.slideshare.net/johnryanrizal/chapter15-a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7*

The buoyant force is due to the displaced fluid. The block material doesn't matter.

CALCULATING BUOYANT FORCE

The buoyant force FB is due to the difference of pressure DP between the top and bottom surfaces of the submerged block.

$$\Delta P = \frac{F_B}{A} = P_2 - P_1; \;\; F_B = A(P_2 - P_1)$$

$$F_B = A(P_2 - P_1) = A(\rho_f g h_2 - \rho_f g h_1)$$

$$F_B = (\rho_f g) A (h_2 - h_1)$$

$$\boldsymbol{F_B = (\rho_f g) V}$$

V is volume of fluid displaced.

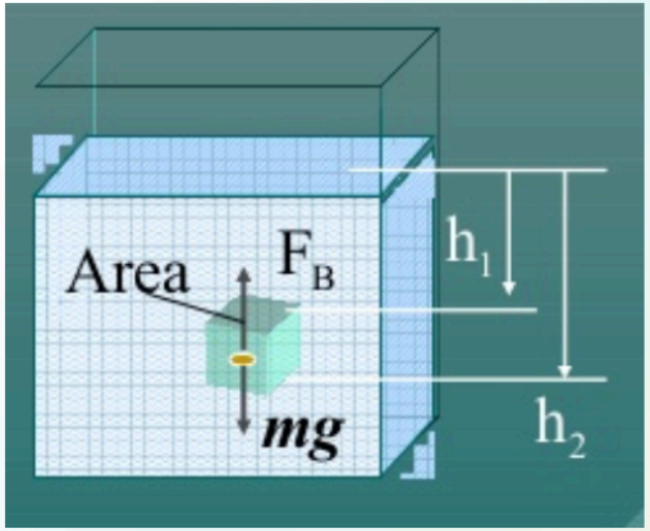


Figure 4. Buoyant Force. Calculation From. *https://www.slideshare.net/johnryanrizal/chapter15-a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7*

Buoyant Force:

$$\boldsymbol{F_B = (\rho_f g) V}$$

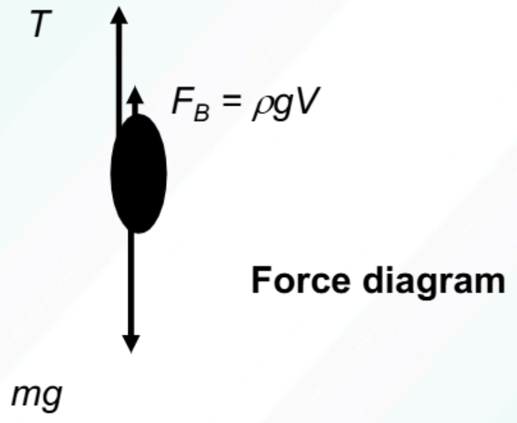
Sample Problems

1. A 2-kg brass block is attached to a string and submerged underwater. Find the buoyant force and the tension in the rope

All forces are balanced:

$F_B + T = mg$

$F_B = \rho_w g V_w$



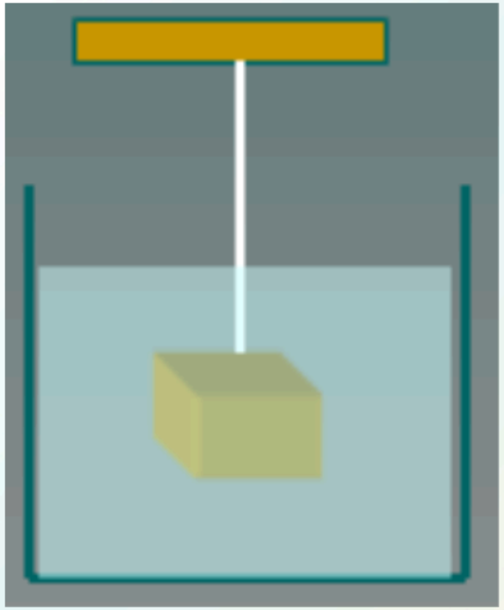


Figure 5. Buoyant Force. Sample From. *https://www.slideshare.net/johnryanrizal/chapter15-a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7*

FLOATING OBJECTS

When an object floats, partially submerged, the buoyant force exactly balances the weight of the object.

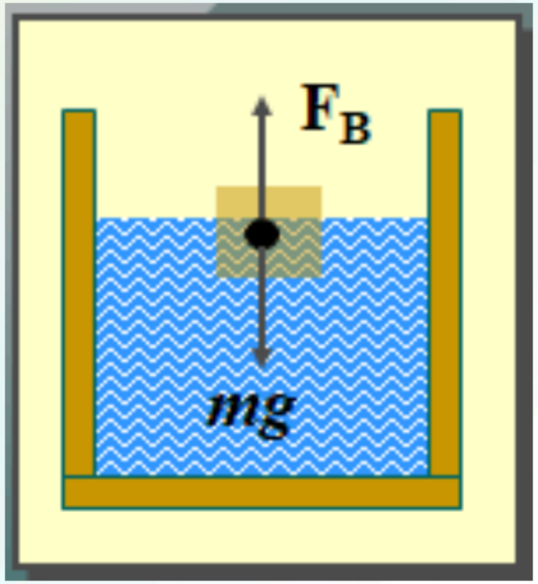


*Figure 6.*Floating Objects. From. *https://www.slideshare.net/johnryanrizal/chapter15-a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7*

$$F_B = \rho_f g V_f \qquad m_x g = \rho_x V_x g$$

$$\rho_f \cancel{g} V_f = \rho_x V_x \cancel{g}$$

Floating Objects: $\boxed{r_f V_f = r_x V_x}$

If V_f is volume of displaced water V_{wd}, the relative density of an object x is given by:

Relative Density:

$$\rho_r = \frac{\rho_x}{\rho_w} = \frac{V_{wd}}{V_x}$$

Sample Problems

2. A student floats in a salt lake with one-third of his body above the surface. If the density of his body is 970 kg/m³, what is the density of the lake water?

Assume the student's volume is 3 m³.

$V_s = 3\ \mathrm{m}^3;\ \ V_{wd} = 2\ \mathrm{m}^3;\ \ r_s = 970\ \mathrm{kg/m}^3$

$$\frac{\rho_s}{\rho_w} = \frac{V_{wd}}{V_s} = \frac{2\ \mathrm{m}^3}{3\ \mathrm{m}^3}; \quad \rho_w = \frac{3\rho_s}{2}$$

$$\rho_w = \frac{3\rho_s}{2} = \frac{3(970\ \mathrm{kg/m}^3)}{2}$$

$$\rho_w = 1460\ \mathrm{kg/m3}$$

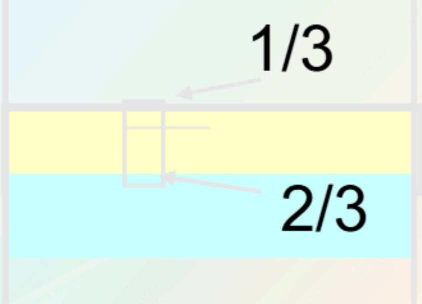


*Figure 7.*Floating Objects. Sample From. *https://www.slideshare.net/johnryanrizal/chapter15-a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7*

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Physics book by Gary B. Conrado and Maria A. Boncodin.

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Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

REFERENCES

https://www.slideshare.net/nosuhaila/chapter2-24683465

https://www.slideshare.net/johnryanrizal/chapter15a?qid=783b9047-7f59-405c-8a1e-629327afd5ba&v=&b=&from_search=7